Curriculum Vitae

Ashish Saxena

Phone: 9599858995
E- mail: ashishsaxenaemailid@gmail.com

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PROFILE SUMMARY

- Good Knowledge of C, Golang, Data Structures and algorithms.
- Sound Linux background with experience in Datacom and IP security domain.
- Good programming experience with debugging skill.
- Ability to work in disciplined & multifunctional team with responsibility.
- Deep knowledge of L2-L3 level networking protocols.
- Experience in developing, Deploying and porting applications in containers on Kubernetes platform.

EDUCATIONAL PROFILE

Examination	Institute	Board/ University	Year Passed	Marks %
B.E (Electronics & Communication)	Birla Institute of Technology, Mesra	Birla Institute of Technology, Mesra	2012	73.10
12th	Kendriya Vidhyalaya-2, Jaipur	CBSE	2007	82.20
10th	Tagore Public School, Jaipur	CBSE	2005	85.60

TECHNICAL EXPERTISE

Operating Systems	Linux
Debuggers	GNU debugger (GDB)
Development Environment/IDE	Linux, Vim, Git
Protocols	TCP/IP, Openflow, Netconf
Programming Language	C, Golang, Python
Development/Productivity Tools	Kubernetes, DPDK, VPP
Software Engineering Tools	Wireshark, Qemu/ KVM, Snort, Docker, Container
Programming Skills	Socket programming, Multithreading, Linux IPC, Linux Device Drivers, IPSec
Technical Expertise	Development of network applications for L2 and L3 layers in Datacom, security protocols and SDN/NFV domain. Development and Deploying containerized apps on Kubernetes
Responsibilities	Designing, Coding, Bug Fixing, Module Testing

INDUSTRIAL EXPERIENCE

- **F5 Networks (Nov 2021 – Present)**

  Profile : **Engineer III**
  Technology : C, Golang, Kubernetes, Linux

- **HSC, Gurgaon (Sep 2018 – Nov 2021)**

  Profile : **Principal Engineer**
  Technology : C, Linux

- **ARICENT, Gurgaon (Jan 2018 – Aug 2018)**

  Profile : **Senior Software Engineer**
  Technology : C, Linux

- **VNL, Gurgaon (April 2017- Jan 2018)**

  Profile : **Senior Engineer**
  Technology : C, Linux

- **IOB, Chennai (Oct -2015 – March 2017)**

  Profile : **Assistant Manager (Non-Technical)**

- **ARICENT, Gurgaon (Aug 2012- Sep 2015)**

  Profile : **Senior Software Engineer**
  Technology : C, Linux

PROJECT DETAILS

1. **BIG-IP**

   NOV 2021 – Now

   Project **: BIG-IP Control plane**

   Role **: Software Development**

   Language**: C, Golang**

   Environment & Tools**: Linux, Kubernetes, Containers**

   Responsibilities:

   - Understanding of F5's BIGIP control plane.
   - Development of applications on Kuberbetes using Golang.
   - Deployment of apps in containerized environment.
   - Bug fixing and testing.

2. **SDN-NFV-APP**

   July 2020 – Now

Project **: SDN-NFV-APP**

Role **: Software Development**

Language**: C**

Environment & Tools**: Linux, Git,OpenNESS , docker, DPDK, VPP,Netconf, libyang, FRR, sysrepo**

Responsibilities:

- Development of application for setting srv6 policies in VPP using NETCONF protocol.
- Integration of router plugins with VPP using FRR.
- Deployment of Edge Computing platform Openness, creation of Network architecture for running end-to-end use case, creating tunnels between datacenter servers and OpenNESS using VPP.

3. Trust Point Monitor Client

Feb 2019 – July 2020

Project **: Trust Point Monitor Client**

Role **: Software Development**

Language**: C**

Environment & Tools**: Linux, Git, Kafka, RabbitMQ, SSL/TLS, IPSec, librdkafka, libjson, Kernel module programming**

Responsibilities:

- Writing multithreaded network application for getting system events (file and network events) from kernel modules and converting them into JSON format.
- Involved in development of module for sending messages to the message bus (Kafka).
- Securing the communication between monitor client application and message bus server by using IPsec
- Write Linux kernel modules for applying security policy on Linux machines.

4. CWLC for LG TV

Sep 2018 – Feb 2019

Project **: WLC for LG TV**

Role **: Software Development**

Language**: C, Python**

Environment & Tools**: Linux, Git, RESTful api, libcurl, ulfius, DPDK,L2-L3 protocols**

Responsibilities:

- Involved in development of control plane and management plane for Linux based application.
- Made contributions to the development of backend code for interfacing with GUI and maintaining Database for backend.
- Managed the shared memory segment between the fast path and control plane application.
- Development of Datapath application for forwarding packets.

5. VectaStar

Jan 2018 – Sep 2018

Project **: VectaStar**

Role **: Software Development**

Language**: C, python**

Environment & Tools**: Linux, Git, Wireshark, Dockers, ONOS, SDN**

Responsibilities:

- Virtualization of system hardware on docker-containers for large-scale deployment.
- Coding, Documentation, Review, Unit testing.
- Involved in POC on SDN (OpenCORD – ONOS integration).

6. R2-R3 Rural BTS

March 2017 – Jan 2018

Project **: R2-R3 Rural BTS**

Role**: Software Development**

Language **: C**

Environment & Tools**: Linux, SVN, Eclipse, socket programming, multithreading, Linux IPC, Linux kernel programming**

Responsibilities:

- Feature enhancement and Bug fixing.
- Coding, Documentation, Review, Unit testing.
- Worked on Master-Slave BTS configuration and Watchdog implementation.

7. FPA (Fastpath Accelerator)

Nov 2013 – Sep 2015

Project **: FPA (Fastpath Accelerator)**

Role **: Software Development**

Language**: C**

Environment & Tools**: Linux, SVN, Wireshark, OVS, snort, qemu, KVM, Intel DPDK, Openflow,**

Responsibilities:

- Understanding of Linux kernel networking stack.
- Development of **Routing** and **firewall** functionality for fastpath (FPA).
- Understanding of application development using Intel DPDK.
- Feature Development of **L2-L3** and **OVS-FPA (Openvswitch)**.
- Involved in integration of **Aricent ISS (control plane)** stack with fast path.
- Coding, Documentation, Code Review, Unit testing.
- LLD and HLD design from specifications
- Experience in development of network applications**.**

8. GSM BTS (GF1 TRS Fastpath)

Dec 2012 – Nov 2013

Project **: GSM FSM BTS (GF1 TRS Fastpath)**

Role**: Software Development**

Language**: C**

Environment & Tools**: Linux, putty, Clearcase, Cygvin, Linux IPC, multithreading, wireshark**

Responsibilities:

- Feature enhancement and working with the client team members.
- Coding, Documentation, Review, Unit testing.

- Deep understanding of code responsible for voice and data calls setup.
- Performed bug fixing reported from client side.

ACHIEVEMENTS

- Strong coding skills in C, DS
- Represented the school and college in various quiz competitions and won several prizes
- Won the prize for best project in Nalanda training (Aricent)
- Won multiple awards for contribution towards the project